

Whole-field, high-resolution Fourier ptychography with neural pupil engineering

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Fourier ptychographic microscopy (FPM) is a computational imaging technique that achieves high-resolution complex amplitude reconstruction across a large field of view. However, conventional FPM is fundamentally limited to regions near the optical axis due to violations of the shift-invariance assumption in off-axis areas, resulting in challenging edge-of-field-of-view reconstructions. We propose neural pupil engineering FPM, termed NePE-FPM, a physical model that dynamically shifts the pupil function position during reconstruction instead of fixing the pupil function in the center of the pupil plane. NePE-FPM engineers the pupil function using an implicit neural representation with multi-resolution hash encoding, enabling continuous, smooth shifting of the pupil function without introducing additional physical parameters. By optimizing a feature-domain loss function, NePE-FPM adaptively filters Fourier-space information from low-resolution measurements, achieving accurate off-axis reconstruction without modeling off-axis propagation. Experimental results demonstrate isotropic resolution of 1149 lp/mm across an 11 mm² FOV using a 4 × /0.1 NA objective. The NePE-FPM bridges the gap between theoretical FPM capabilities and practical whole-slide imaging demands. © 2025 Optica Publishing Group under the terms of the [Optica Open Access Publishing Agreement](#)

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1. INTRODUCTION

Fourier ptychographic microscopy (FPM) is a powerful computational imaging technique that combines the principles of synthetic aperture and phase retrieval to achieve combinations of improved resolution and extended field of view (FOV) in optical microscopy [1,2]. Since its initial demonstration in 2013, FPM has revolutionized quantitative phase imaging by enabling gigapixel-scale reconstructions with resolution exceeding the diffraction limit of the objective lens. FPM's remarkable capability stems from FPM's use of angularly varying illumination to sequentially capture multiple low-resolution intensity images, which are then computationally synthesized into a high-resolution complex amplitude reconstruction through iterative phase retrieval algorithms [1,3–5]. The technique's unique ability to simultaneously achieve high numerical aperture (NA) and wide FOV has found applications across diverse fields, including digital pathology [1,6,7], where FPM enables whole-slide imaging with subcellular detail; live cell analysis [8,9], where FPM's label-free quantitative phase imaging capability provides non-invasive monitoring of cellular dynamics; and industrial inspection [10–12], where FPM achieves large FOV and high resolution that are critical for defect detection. Moreover, FPM's compatibility with standard microscope hardware has facilitated its rapid adoption, as FPM's illumination scheme can be implemented as an add-on module to existing systems with minimal modifications. The concept of FPM's phase retrieval paradigm has inspired extensive variations of the phase retrieval framework, with enhanced quantitative phase imaging

ability. Without changing the illumination scheme, the FPM is able to achieve optical tomography [9,13–16], digital adaptive optics phase imaging [17,18], and lens-free imaging [19]. FPM can be extended to far-field detection, achieving remote sensing [20,21] with enhanced resolutions.

Despite the aforementioned advantages, conventional FPM implementations face a fundamental limitation that has constrained FPM's practical utility: the technique's performance degrades substantially for off-axis regions of the FOV [2,22]. While FPM theoretically promises uniform resolution across its entire synthetic aperture, in practice, researchers have consistently observed that high-quality reconstructions are typically limited to a small area near the optical center [23–25]. This discrepancy between theory and practice represents one of the most significant challenges preventing FPM from realizing its full potential in real-world applications, particularly in clinical and industrial settings where consistent performance across large FOVs is essential [26,27].

The aforementioned limitation originates from the violation of shift-invariance assumptions in optical systems [24]. In ideal imaging scenarios, the illumination fields are plane waves that uniformly shift the sample's Fourier spectrum across the entire FOV. The point spread function remains constant across the entire FOV, a condition that holds reasonably well for small areas near the optical axis. However, when moving toward the edge of FOV, optical aberrations become increasingly pronounced, and the system's transfer function exhibits noticeable spatial variations [23,24]. The finite distance between the LED dots and sample leads to a

spherical illumination wavefront, further breaking the approximation of the conventional FPM model. Several approaches have been proposed to address these challenges. Early attempts focused on segmenting the FOV into smaller regions and applying position-dependent aberration corrections [1,28]. While somewhat effective, these methods significantly increase computational complexity and require careful calibration procedures [29]. More recent efforts have explored using Zernike polynomials to model spatially varying aberrations [30] or develop an alternative FPM solver based on a closed-form solution [31], but these approaches are limited by their inability to capture high-frequency wavefront variations. Creating a complex imaging forward model of FPM by taking the wavefront curvature of the illumination into consideration can suppress the artifact of off-axis reconstruction [24,25,32]; however, these methods will introduce additional system parameters that cannot be precisely obtained. Developing a specified optical layout for LED illumination can address the off-axis problem of conventional FPM [33], but requires customized optical elements.

Revisiting the off-axis reconstruction problem, the core challenge lies in the mismatch between the conventional FPM model and the actual imaging physics. While previous works have used full-wave Fresnel diffraction to approximate image formation across the field of view (FOV) [23,32], this approach requires precise system parameters and still remains approximate due to the non-negligible LED size. Notably, we observed that reconstructed Fourier spectra in off-axis regions exhibit slight frequency shifts—equivalent to displacements of the pupil function in the pupil plane. Motivated by this, we propose a neural pupil engineering strategy, where the pupil function is no longer fixed but dynamically shifts across the pupil plane in a spatially adaptive manner to better reflect the true image formation physics [34–36]. In this work, we introduce a fundamentally different scheme for whole-field FPM reconstruction, termed neural Pupil Engineering FPM (NePE-FPM). The NePE-FPM dynamically models the pupil function's spatial position throughout the reconstruction process. The shifting pupil function allows us to capture and compensate for spatially varying optical transfer functions while strictly adhering to the physical principles of Fourier ptychographic microscopy. Our key innovation is representing the pupil function with an implicit neural representation powered by multi-resolution hash encoding, which enables a continuous and smooth adaptation to off-axis frequency coordinates, without introducing additional physical parameters. NePE-FPM delivers several critical advantages over existing methods. First, NePE-FPM embeds physical modeling within a flexible computational framework, offering accurate correction for off-axis aberrations, and addressing phase curvature of LED's spherical wave illumination. Second, the use of hash encoding allows for a compact yet expressive representation of spatially varying pupil behaviors with minimal computational overhead. Third, the NePE-FPM retains the inherent parallelism of FPM, supporting efficient and scalable implementation on modern GPU architectures. By integrating physics-driven modeling with learning-based representations, NePE-FPM establishes a new direction for robust and efficient off-axis reconstruction, offering a generalizable foundation for future advances in computational microscopy.

2. RESULTS

A. Neural Pupil Engineering

The experimental setup of an FPM system is shown in Fig. 1(a), which consists of a programmable LED array illuminating the sample at varying angles, with light collected through a $4f$ imaging system and captured by a camera. The LED array produces angular diversity in illumination and enables frequency-domain coverage beyond the conventional limit imposed by the objective's numerical aperture (NA). A low-resolution image is captured when an individual LED is turned on. After all LEDs are used, a data cube that encodes high-frequency information beyond the NA boundary is collected for FPM reconstruction.

The image formation model of noise-free FPM is written as [1]

$$\mathbf{I}_n = |\mathbf{F}^\dagger \mathbf{P}_n \mathbf{F} \mathbf{U}|^2, \quad (1)$$

where $\mathbf{I}_n$ denotes the intensity measurement illuminated by n th LED. $\mathbf{F}$ and $\mathbf{F}^\dagger$ denote forward and inverse Fourier transforms. $\mathbf{P}$ denotes the pupil function. $\mathbf{M}_n$ denotes the selective matrix of n th LED, and $\mathbf{U}$ is the complex amplitude of sample.

In traditional FPM, the pupil function is constrained by the coherent transfer function (CTF) and is typically assumed to be a fixed, centered circular binary mask in Fourier space [30,37,38], located at $f_x = f_y = 0$, where f_x, f_y denote spatial frequency coordinates. This simplification overlooks the spatially varying nature of image formation, especially under off-axis illumination. To overcome the limitation, we digitally engineer the pupil function, which is no longer statically defined but is instead treated as a dynamic, learnable optical element that adapts across the field of view. As illustrated in Fig. 1(b), we model the pupil function as a continuously drifting entity in Fourier space, capturing the frequency shift of the synthetic aperture under off-axis illumination. The core of this neural pupil engineering framework is an implicit neural representation of the pupil function, as shown in Fig. 1(c). Here, frequency coordinates are mapped to complex-valued optical responses using a learnable multilayer perceptron (MLP), augmented by a hash table that encodes spatial information through learnable embedding vectors. The MLP outputs both amplitude, $|\mathbf{P}(f_x, f_y)|$, and phase, $\angle \mathbf{P}(f_x, f_y)$, enabling the pupil function to shift during reconstruction and adaptively match the off-axis optical behavior. The NePE-FPM integrates the physical forward model with data-driven optimization, enabling high-fidelity reconstruction of off-axis areas without intricate modeling of off-axis beam propagation in a $4f$ imaging system [39].

The positional encoding is the key for implicit neural representation for pupil function [40]. Unlike using sinusoidal [40] or tensor decomposition [41,42] for positional encoding, we chose a multi-resolution hash table for positional encoding, which is inspired by instant neural graphic primitive (Instant-NGP) [43] as shown in Fig. 1(c). The multi-resolution hash table stores trainable embedding vectors as latent positional encoding. A tiny MLP is trained, in the meantime, to decode the embedding vectors and generate the final output of positional encoding. In the multi-resolution hash table, coarse hash grids capture low frequencies of the pupil function, while fine grids resolve high frequencies of the pupil function [43]. The multi-resolution hash table naturally supports sub-pixel precision and embeds physical constraints such as the positional boundary of the pupil function without explicit regularization. Moreover, its $O(1)$ query complexity and memory efficiency enable real-time optimization across millions of pixels,

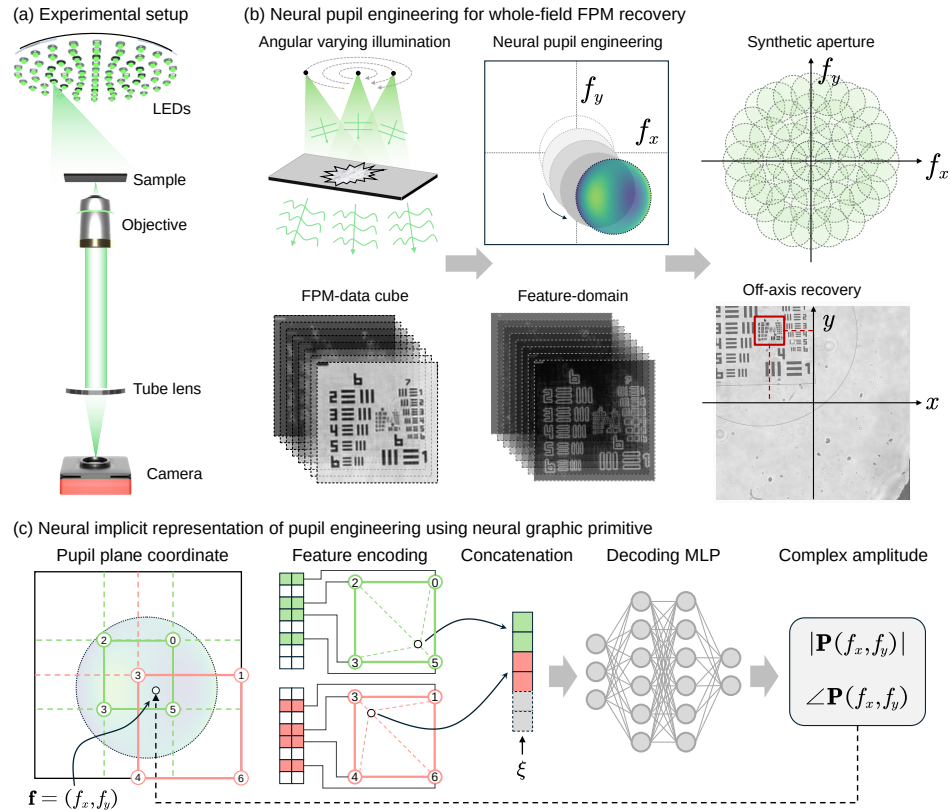


Fig. 1. Neural pupil engineering FPM pipeline. (a) Optical setup for FPM experiments. (b) Scheme of NePE-FPM reconstruction. The pupil function moves in the Fourier plane as the iteration progresses. (c) Scheme of neural pupil engineering, where the pupil function is represented using a paired multi-resolution hash table and an MLP. ξ denotes the embedding vector from other levels of hash table.

a critical requirement for large-scale experimental data in FPM. To further accelerate the positional encoding, we write highly optimized CUDA code for large-scale parallel execution.

The implicit neural representation of pupil function is merged into the FPM reconstruction. Here we use the feature-domain FPM loss function for training the sample's complex amplitude and the parameters of the implicit neural representation, denoted by θ . The feature-domain loss function is given as [6,44]

$$\mathcal{L}(\mathbf{U}, \theta) = \left\| \nabla \sqrt{\mathbf{I}_n} - \nabla |\mathbf{F}^\dagger \mathbf{P}(f_x, f_y; \theta) \mathbf{M}_n \mathbf{F} \mathbf{U}| \right\|_1. \quad (2)$$

We take advantage of auto-differentiable programming [38,45], but in a different way. Instead of relying on the default automatic differentiation routines of MATLAB, we implement customized CUDA programs to explicitly compute the forward and backward passes of FPM with minimized computational redundancy. These CUDA programs are wrapped as differentiable functions in MATLAB, enabling the efficient CUDA computation pipeline to be seamlessly integrated into MATLAB's auto-differentiation framework for fast and flexible FPM reconstruction.

B. Dynamic Drifting of Pupil Function

Figure 2 shows the reconstruction of plant root tissue using fixed pupil models (FD-FPM) and drifting pupil models (NePE-FPM). Figure 2(a) shows the plant root slice, and Fig. 2(b1) presents the FD-FPM reconstruction for the entire field view, with the two red

box regions magnified in Figs. 2(b2) and 2(b3) to highlight unresolved high-frequency details. The off-axis reconstruction results are shown in Figs. 2(c) and 2(d), where Figs. 2(c2) and 2(d2) show the recovered pupil amplitude, revealing a static circular support (dashed line) constrained by the NA. The corresponding sample reconstruction in Figs. 2(c1) and 2(d1) shows blurred cellular structures.

On the contrary, Figs. 2(c3) and 2(d3) demonstrate the neural-implicit recovered pupil amplitude, exhibiting a shifted pupil position toward the upper-right direction and upper-left direction, with significant shifting distance. Since the reconstruction quality is highly sensitive to the aperture's spectral position, only the correct aperture placement can yield satisfactory reconstruction results. Thus, the pupil shifting reconstruction enables NePE-FPM to resolve finer structures as shown in Figs. 2(c3) and 2(d3), especially in Fig. 2(c3), where each individual cell, stained brown, can be resolved. Figures 2(e1) and 2(e2) plot the intensity profile along the line l_1 and l_2 in (c1) and (d1), where the intensity profile of the raw image varies smoothly due to the loss of high resolution. While both FD-FPM and NePE-FPM obtain resolution-enhanced reconstruction, and fine cellular structures can be resolved. The contrast of the NePE-FPM is better than that of the FD-FPM, as seen from the green curves in Figs. 2(e1) and 2(e2) within $x \in [40 \mu\text{m}, 50 \mu\text{m}]$. The image contrast is further quantitatively evaluated using multi-level image contrast [46] as shown in Fig. 2(f), where the NePE-FPM achieves about 55% enhancement in image contrast for Fig. 2(c1) and about 12% enhancement for Fig. 2(d1). Reconstruction of the central FOV is available in Section 1 of the Supplement 1.

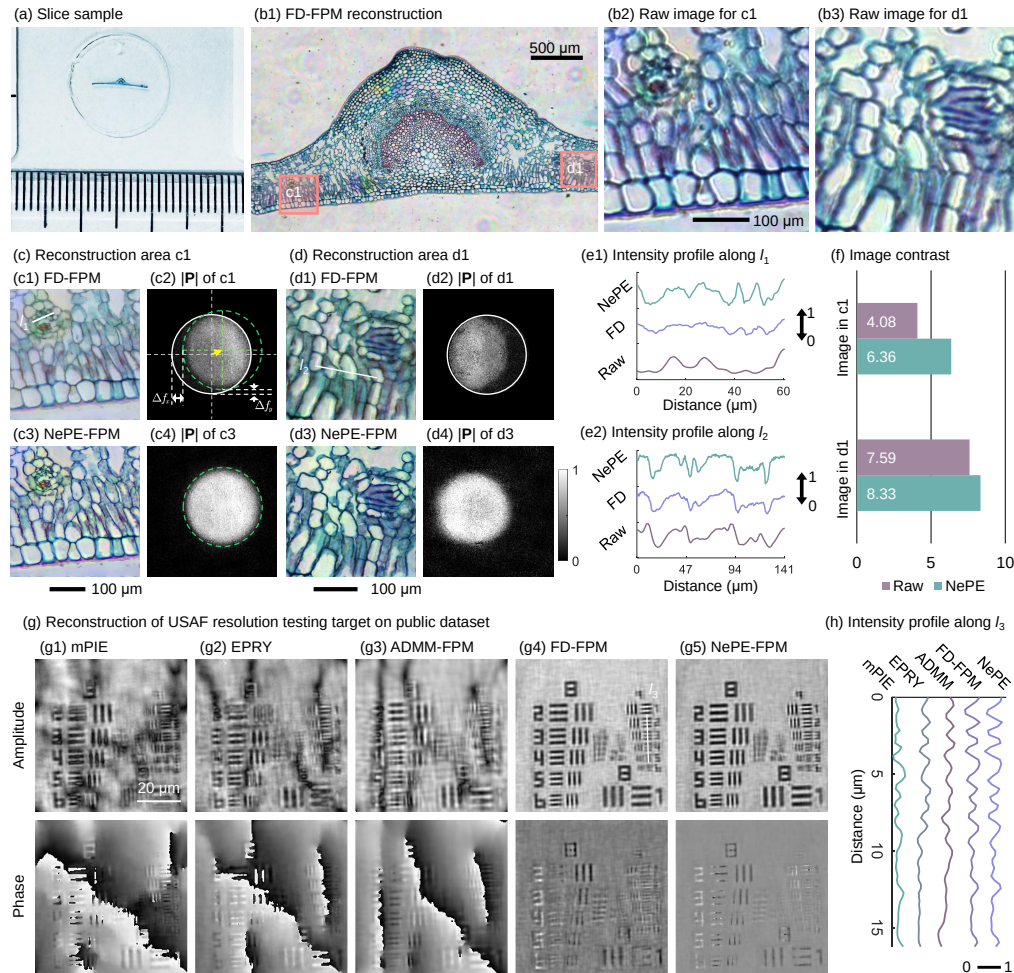


Fig. 2. FPM reconstruction of a fruit root slice. (a) Slice sample. (b1) Full-field FPM reconstruction using FD-FPM. (b2) and (b3) Raw image in the area marked in the red box marked with c1 and d1 in (b). (c1) Reconstruction results using FD-FPM. (c2) Amplitude of pupil function in green channel. (c3) Reconstruction results using NePE-FPM. (c4) Amplitude of pupil function in green channel. (d) Raw image in the area marked in the red box marked with d1 in (b). (d1) Reconstruction results using FD-FPM. (d2) Amplitude of pupil function in green channel. (d3) Reconstruction results using NePE-FPM. (d4) Amplitude of pupil function in green channel. (e1) and (e2) Intensity profiles along lines l_1 in (c1) and l_2 in (d1). (f) Multi-level image contrast [46]. (g) Reconstruction of USAF resolution testing target on public dataset from [24], (g1)–(g5) are results from 5 FPM algorithms. (h) Intensity profile along the line l_3 marked in (f4).

An interesting observation is that the recovered pupil function in FD-FPM also shifts away from the center but becomes truncated by the CTF. This suggests that the CTF constraint may degrade reconstruction quality and is not always necessary to apply during reconstruction. However, additional experiments in Section 3 of the Supplement 1 indicate that the CTF constraint in fact facilitates convergence in FPM reconstruction. Without enforcing the CTF constraint, the reconstructed pupil function tends to be underestimated although it is shifted as well, leading to a substantial decline in overall reconstruction quality.

Figure 2(g) shows the reconstruction results of the USAF resolution testing target from a public dataset in [24], dedicated to addressing the phase curvature in Fourier ptychography. The NePE-FPM is compared against four other methods, namely the ptychographic iterative engine with momentum (m-FPM) [47], FPM with embedded pupil function recovery (EPRY) [37], FPM with alternating direction method of multipliers (ADMM-FPM) [48], and conventional feature-domain FPM [6,44]. The conventional FPM methods as given in Figs. 2(g1)–2(g3) fail to reconstruct the amplitude and phase of the sample due to the phase

curvature of the LED illumination, corresponding to the experimental results given in [24]. While the feature-domain methods successfully recover the sample's amplitude and phase as shown in Figs. 2(g4) and 2(g5), as can also be seen from Fig. 2(h), where the amplitude profile of NePE-FPM has better resolution and higher image contrast. Within the NePE-FPM, a phase pattern with better visual quality was obtained, where the stripes of the resolution target are visible. Notably, as the NePE-FPM uses the feature-domain loss function for training, the reconstructed amplitude in Fig. 2(g5) is similar to that of the FD-FPM in Fig. 2(g4). But the quality of the reconstructed phase pattern of NePE-FPM is better than that of the FD-FPM, since the phase is more sensitive to the pupil function than that of the amplitude.

The LED illumination wave is actually a spherical wave due to the finite distance between the LED panel and the sample. When the sample is illuminated by the spherical wave, the amount of frequency shift at different spatial positions is actually different. By introducing pupil shifts in the frequency domain during the iterative reconstruction, the algorithm can adaptively capture and

correct phase curvature, thereby improving consistency between the forward model and the experimentally observed data.

C. Off-Axis FPM Reconstruction

Figure 3 presents a systematic evaluation of the neural-implicit pupil-drifting FPM (NePE-FPM) method through large-field reconstruction and iterative optimization analysis. The full-field reconstruction in Fig. 3(a) demonstrates the conventional FD-FPM result. The zoomed-in view in Fig. 3(b) reveals NePE-FPM's superior resolution compared against the FD-FPM for off-axis reconstruction, enabled by NePE-FPM's expanded frequency coverage evident in the shifted Fourier spectrum as shown in Fig. 3(c).

The spectrum clearly shows additional off-axis components captured by the pupil shifting model, which conventional methods miss due to fixed pupil constraints.

The iterative reconstruction process in Fig. 3(d) illustrates how both sample amplitude and pupil function progressively refine across training epochs. Early iterations (epochs 15–30) primarily correct low-frequency wavefront aberrations, while later stages (epochs 45–75) recover fine cellular details, such as organelle membranes in the highlighted region. This evolution aligns with quantitative metrics in (e). The PSNR and SSIM exhibit critical improvement during epochs 30–45, eventually stabilizing at high-fidelity levels ($\text{SSIM} \geq 0.9$). The convergence behavior suggests

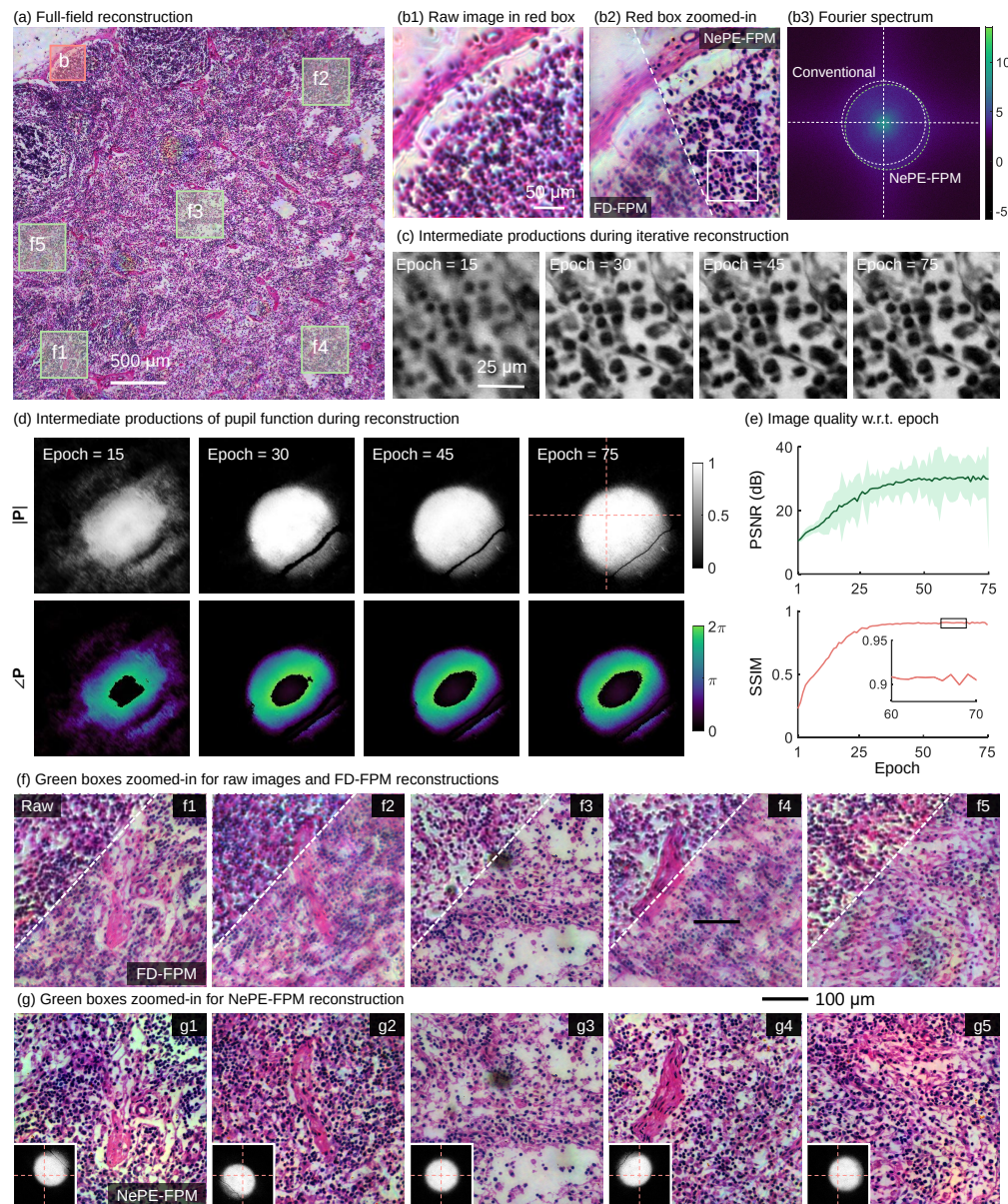


Fig. 3. Off-axis FPM reconstruction over a large field of view. (a) Full-field reconstruction using FD-FPM. (b) Comparison of reconstructed intensity for the area marked in the red box in (a). (c) Shifted Fourier spectrum of (b) reconstructed using NePE-FPM. (d) Intermediate products for the green channel, including sample intensity and pupil function, during NePE-FPM reconstruction for the area marked in the white box in (b). (e) Reconstruction quality, including peak signal-to-noise ratio (PSNR) and structure similarity index measurement (SSIM) with respect to training epochs. Reconstructed images after 200 epoch are the references for the evaluations. (f) Off-axis reconstruction in five areas marked in green in (a) using FD-FPM. (g) Off-axis reconstruction in five areas marked in green in (a) using NePE-FPM.

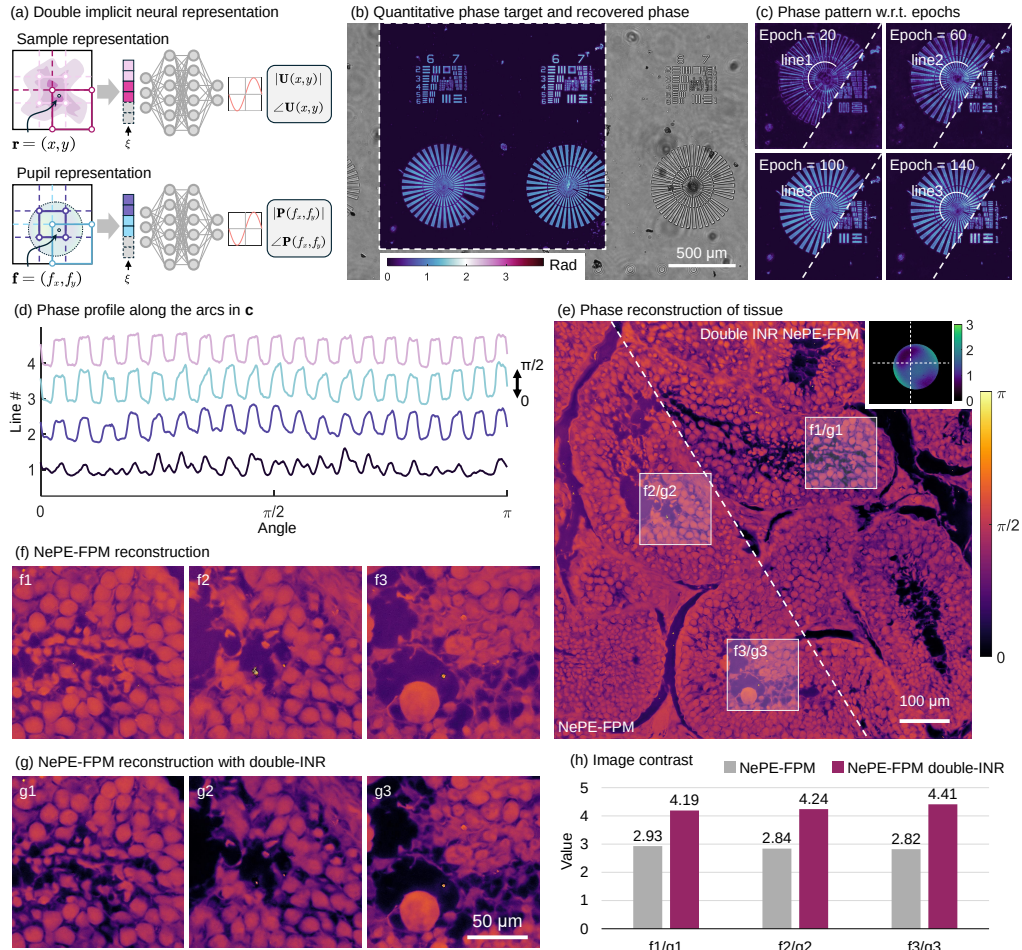


Fig. 4. Quantitative phase imaging of NePE-FPM using double-implicit neural representations. (a) Sketch of double INR. The sinusoidal function is used for the activation function of the outputs. (b) Quantitative phase target and the NePE-FPM reconstruction. (c) Evolution of phase pattern with respect to training epoch. (d) Quantitative phase profile along the white arch in (c). (e) Phase pattern of a tissue slice. The NePE-FPM is combined with FD-FPM. (f) and (g) Phase pattern reconstructed using FD-FPM and NePE-FPM, respectively. (h) Image contrast evaluation for image pairs (f1/g1)–(f3/g3).

a distinct phase transition in the optimization, where the NePE-FPM can achieve initial rapid correction of global distortions, followed by gradual recovery of high-frequency features.

Validation across multiple off-axis areas in Fig. 3(f) confirms the method's robustness, with NePE-FPM consistently resolving structures missed by conventional approaches. Features like vesicle clusters (f1/f3) and membrane edges (f2/f4) are clearly reconstructed. The shifted pupil centers follow FOV-dependent trajectories, where the recovered pupil function tends to shift toward the center of the FOV. The amplitude of the pupil function exhibits vignetting effects as the edge of the amplitude smoothly vanishes. In general, the results in Figs. 3(f) and 3(g) demonstrate that neural-implicit pupil modeling not only overcomes the off-axis challenge in the synthetic aperture of traditional FPM, but also does so through a physically interpretable optimization process that implicitly aligns with wave optics principles, without intricate wave-optics modeling of image formation.

We further test the proposed framework by employing two coordinated neural networks to simultaneously model the sample's complex field $U(x, y)$ and the pupil function $P(f_x, f_y)$, as illustrated in Fig. 4(a). The sample implicit neural representation, parameterized by φ encodes spatial coordinates (x, y) into the amplitude and phase of sample ($|U|$ and $\angle U$), while the

pupil implicit neural representation, parameterized by θ , maps frequency coordinates (f_x, f_y) to the complex aperture response ($|P|$ and $\angle P$). The loss function is given as

$$\mathcal{L}(\varphi, \theta) = \left\| \nabla \sqrt{I_n} - \nabla |F^\dagger P(f_x, f_y; \theta) M_n F U(x, y; \varphi)| \right\|_1. \quad (3)$$

This double representation enables joint optimization of both domains, with sine-based activations ensuring smooth complex continuity, and numerical values' physical boundary [49]. For example, the amplitude of the sample will be no larger than 1, and the wrapped phase is always in $[0, 2\pi]$.

Validation using a quantitative phase target as shown in Fig. 4(b) demonstrates NePE-FPM's accuracy in recovering ground-truth phase patterns. The iterative reconstruction process in Fig. 4(c) shows phase reconstruction evolving from low-frequency shape approximation (10 epochs) to high-fidelity detail recovery (50 epochs), with the phase profile in Fig. 4(d) along the marked arc confirming phase retrieval accuracy. Moreover, we found that the algorithm tries to first recover the high-frequency component of the phase pattern as shown in Fig. 4(c) for epoch ≤ 60 and then refines the low-frequency components of the phase pattern, which corresponds to the nature of FPM since the number of dark-field images is more than that of bright-field images.

Like conventional FPM, the NePE-FPM cannot recover the very low-frequency components of the phase pattern due to the loss of the low-frequency component of the phase transfer function [15,16,50].

Biological application is shown in Fig. 4(e), where the double-implicit neural representations of NePE-FPM reveal distinct phase structures in tissue slices with enhanced contrast compared to conventional NePE-FPM. The background signal is efficiently suppressed, and thus, the contrast of the phase pattern is improved, without loss of phase fidelity. Notably, while both methods incorporate total variation regularization, with the same strength, to suppress noise, the implicit neural representation demonstrates superior artifact-handling capabilities compared to conventional approaches. Conventional NePE-FPM reconstructions [Fig. 4(f)] exhibit characteristic staircase artifacts—piecewise-constant intensity jumps that distort phase gradients—due to the L_1 -norm's tendency to over-promote sparsity in discrete representations [51]. In contrast, the double-implicit neural representation of NePE-FPM, in Fig. 4(g), inherently avoids such artifacts through its continuous function parameterization. The coordinate-based MLPs act as a spectral filter, naturally smoothing high-frequency noise while preserving genuine phase transitions, evidenced by the smooth profiles in Fig. 4(d). Image contrast evaluation in Fig. 4(h) further indicates the improvement of the phase reconstruction using NePE-FPM.

D. Computational Efficiency

The framework demonstrates robust phase reconstruction under white-light LED illumination as shown in Fig. 5, where the sample

is illuminated by a white-light LED, and the images were captured by a monochromatic camera. The NePE-FPM successfully recovers high-resolution structures of the phase target from the raw data in Fig. 5(a2), outperforming conventional FPM [37] in Fig. 5(a2) in both clarity and noise suppression. Wavelength-specific reconstructions in Figs. 5(b1)–5(b3) reveal consistent performance across different spectral bands (red: 660 nm, green: 532 nm, and blue: 488 nm), with the phase profiles in Fig. 5(c) showing <5% deviation in optical path length measurements between colors. NePE-FPM achieves spectral stability by implicitly separating wavelength-dependent aberrations during optimization. Additional experiments are available in Section 4 of the Supplement 1.

Computationally, the NePE-FPM achieves fast processing through two engineering innovations: (1) The hash-accelerated neural pupil representation, which CUDA acceleration reduces inference time to ≤ 100 ms per 4 million of coordinates queries for an image of 2024×2048 pixels as shown in Fig. 5(d). While the custom CUDA pipeline processes a raw FPM data cube of $2048 \times 2048 \times 93$ pixels in less than 25 s for 20 epochs, with $\times 8$ upsampling rate, which is $15\times$ speedup over MATLAB/CPU implementations. The plotted scaling behavior confirms near-linear complexity with image size, critical for large-FOV applications. For raw data of $512 \times 512 \times 93$ pixels, the execution time for 20 epochs is less than 1 s. Details for the CUDA-accelerated FPM are available in Section 6 of the Supplement 1.

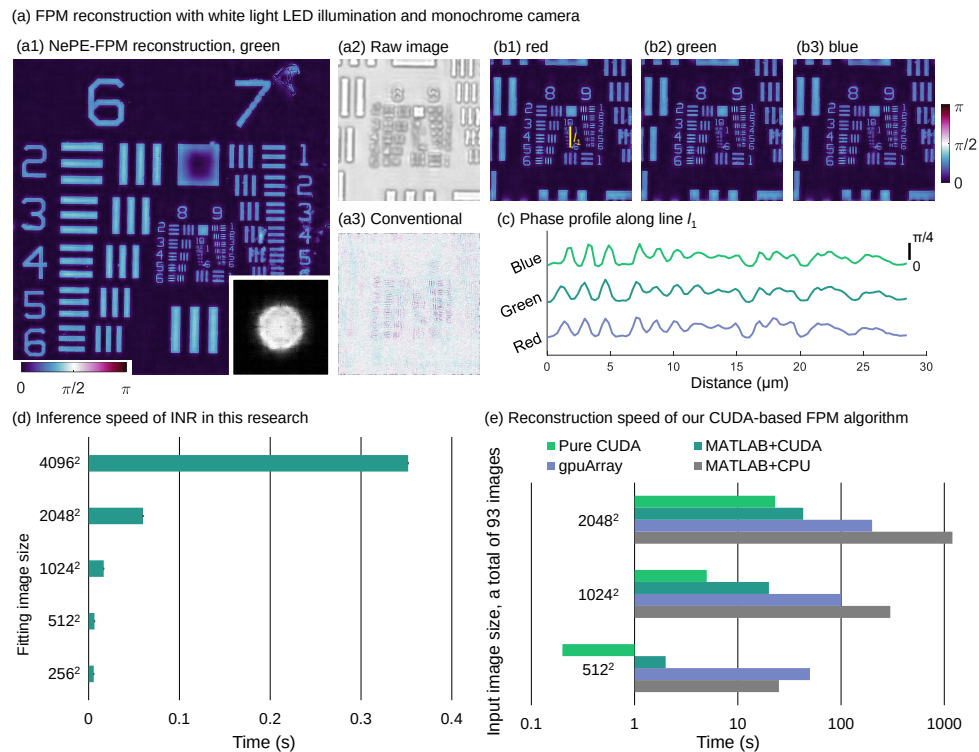


Fig. 5. Reconstruction efficiency of NePE-FPM. (a) FPM phase reconstruction with white light LED illumination and monochrome camera. (a1) Phase pattern. (a2) Raw image for groups 8–10. (a3) Reconstruction results using the conventional FPM algorithm. (b) Reconstruction of phase pattern with different wavelength parameters. (b1)–(b3) results for red (0.660 μm), green (0.532 μm), and blue (0.488 μm) wavelength. (c) Quantitative phase profile along the yellow line in (b1). (d) Computational efficiency of hash-NPG in this research. (e) Computational efficiency of our developed CUDA-accelerated FPM algorithm. All algorithms ran for 20 epochs with a batch size of 25 images.

3. DISCUSSION

This study presents a comprehensive neural-implicit framework for Fourier ptychographic microscopy (FPM), addressing key challenges in off-axis synthetic aperture through shifting the pupil function in Fourier space in optical modeling of the forward model of FPM. Our neural pupil engineering FPM (NePE-FPM), powered by implicit neural representations, overcomes the synthetic aperture limit of conventional FPM by learning continuous, physics-compliant pupil functions. The position of the pupil function is no longer restricted to the center of the Fourier space; instead, the pupil function can adaptively shift in the Fourier space to find the position that can produce the FPM reconstruction results with the best quality. The multi-resolution hash encoding enables adaptive frequency-band allocation, demonstrates its biological impact, resolving cellular structures beyond the diffraction barrier of the objective lens, and a linear shifting invariant model with quantifiable edge sharpness improvements. The recovered pupil shifts follow FOV-dependent geometry, and the pupil function is not allowed to shift beyond the area of the pupil plane; otherwise no information can pass through the pupil leading to the failure of reconstruction.

The double-implicit neural representation architecture unifies sample and pupil optimization, though with longer reconstruction duration, achieving aberration-free phase imaging with reduced discrete artifacts. As evidenced in Fig. 4, this approach uniquely combines noise suppression with artifact-free reconstruction, leveraging implicit neural representations' implicit continuity to avoid staircase effects plaguing conventional methods. Another advantage of double-implicit neural representation is its decoupling of target amplitude and phase, so that they can be separated and modified with different non-linear activation functions, with specific boundaries. For example, the amplitude of the sample is a positive number and will not exceed to 1. Simply using a sine function with shifted minima for the amplitude output of the network can achieve the amplitude nature without explicitly introducing regularization terms. Furthermore, by replacing the pixel-wise representation of the sample with a learnable hash table and MLP, the landscape of the loss function can be smooth, benefiting the algorithmic convergence [52].

Notably, the computational pipeline of FPM reconstruction reduces reconstruction time to 0.3 s for 4 MP images. Combined with the CUDA-accelerated hash table query, the reconstruction time for raw images of 4 MP takes about 30 s. The CUDA-accelerated computational pipeline represents a pivotal engineering breakthrough that transforms neural-implicit FPM from a theoretical concept into a practical imaging solution. By deeply optimizing the reconstruction workflow using kernel fusion for GPU architectures, our implementation achieves unprecedented processing speeds while maintaining rigorous algorithmic accuracy [53]. The hierarchical parallelization strategy simultaneously exploits coarse-grained parallelism across LED illumination patterns and fine-grained pixel-level parallelism in neural inference and Fourier operations, reducing latency by 92% compared to conventional GPU offloading approaches [54]. This efficient computational design enables sub-second reconstruction of 4 MP images for conventional FPM reconstruction, making high-throughput quantitative phase imaging clinically viable.

Collectively, by harmonizing wave optics physics with data-driven optimization, the proposed framework achieves: (1)

high-quality off-axis FPM reconstruction by implicitly modeling the shifting pupil using the combination of neural network and hash table; (2) artifact-free quantitative phase imaging via continuous representations, and (3) scalability to fast and real-time applications via CUDA acceleration. Future work could extend the proposed architecture to whole-slide imaging for pathological analysis [7,55] or integrate it with programmable optics for hardware-in-the-loop optimization. The NePE-FPM can be extended to dynamic scenarios [56,57] by encoding the spatiotemporal complex wavefront into the hash encoding.

While our proposed method achieves significant advances in Fourier ptychographic microscopy, it does present certain limitations that warrant discussion. To address spatially varying aberrations, NePE-FPM, like conventional FPM, still requires cropping the full FOV into smaller patches, reconstructing each independently, and then stitching them together to obtain the final large-FOV image. Differently, the cropped patch size for NePE-FPM can be larger than that for conventional FPM, due to the neural pupil engineering. Second, the current implicit neural representation of the pupil function only captures variations along the spatial frequency coordinates f_x and f_y . The 2D representation can be expanded into 4D, for (x, y, f_x, f_y) . The 4D coordinate would allow the model to account for both spatial and frequency-domain variations, thereby enabling direct handling of spatially varying aberrations across the entire FOV. Third, reconstruction speed for large FoVs is mainly limited by the computational load of the implicit neural representation. We alleviate this bottleneck with an optimized CUDA pipeline that scales nearly linearly with GPU resources. For example, upgrading from an RTX 3090 (10,496 cores) to an RTX 4090 (16,384 cores) is expected to improve speed by 60%. This scalable performance characteristic future-proofs our method against increasingly demanding imaging applications [58,59].

Looking forward, the above limitations actually point toward promising directions for further development. Ongoing advances in neural network compression and specialized hardware acceleration for implicit representations may soon overcome the current speed limitations. The speed of our self-developed hash encoding has not yet reached the fastest theoretical speed of NVIDIA's pure CUDA implementation of instant-NGP [43,60]. Importantly, the method's core advantages, which modeling the movable pupil function with physics-aware neural representation, remain uncompromised.

4. METHOD AND MATERIALS

A. Hardware

All experiments were conducted in our customized Fourier ptychographic platform. The objective lens is of $4\times$, 0.1 NA. The raw images were collected with a monochrome PCO camera of 16 bits, in the same exposure duration. No data pre-processing was applied for FPM reconstruction. The sample was illuminated using a commercial LED panel with a circular-shaped layout (WS2812B 5050 RGB LED, 93 LEDs); the distance between two adjacent LED rings is 12 mm. The distance between the LED panel and the sample is 160 mm for the stained sample. The distance is 90 mm for the quantitative phase target. The central wavelengths of red, green, and blue LEDs are 0.660, 0.532, and 0.488 μm , respectively. All the calculations were performed on a personal desktop with an Intel Core i9-12900K CPU at 3.20 GHz, 32 GB of RAM, and

an NVIDIA RTX 3090 graphics card, 24 GB of RAM, running Windows 10 Pro for Workstations.

B. Software

The code was implemented in MATLAB R2024b with NVIDIA CUDA v12.8. Forward and backward FPM calculations were written in CUDA, decorated with MATLAB's differentiable functions enabling auto-differentiation. All the comparison studies on efficiency were performed in the same computational environment. The NePE-FPM was reconstructed using the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.99$ for the parameters. A total of 25 images are contained in a single mini-batch to enhance the likelihood of escaping local minima [44,61]. The maximum resolution of the multi-resolution hash table is determined by the pixel number of the input raw images (e.g., a 512×512 image yields 512×512 maximum resolution). If the pupil function is assumed to vary slowly in the pupil plane, this resolution can be halved to reduce redundancy in the neural representation. Details for the multi-resolution hash table can be found in Section 6 of the Supplement 1.

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Data availability. Data and MATLAB codes underlying the results presented in this paper are available in Ref [62].

Supplemental document. See Supplement 1 for supporting content.

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